Mini Assignment 3

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*Step 3:*

This analysis aims to identify whether the expression levels of the selected miRNAs vary between male and female patients. Since miRNA expression values are continuous and scaled measurements and gender is a nominal variable consisting of two independent groups, there is a need to apply a statistical method that would compare mean expression levels across groups appropriately.

The Independent Samples t-test is the most common parametric method used in this context because it assesses whether the average expression value significantly differs among male and female subjects. It is a requirement to first verify, before actually applying the test, that several underlying assumptions are met in order for the results to be appropriate and valid.

The first assumption involves the normality of miRNA expression data in each group. In order for the t-test to hold, the distribution of expression levels in both male and female participants for every miRNA under consideration should be approximately normal. This assumption is then checked through the Shapiro-Wilk test, which is a formal statistical test of normality, and Q-Q plots, a graphical examination of how well the observed data follow the distribution of a theoretical normal curve. If large departures from normality are observed, the t-test can be unstable, particularly with small sample sizes.

In addition to the normality assumption, the second assumption comprises homogeneity of variance; this requires that the variation in miRNA expression should be similar between the two groups, males and females. This is tested for by Levene’s Test, which identifies if the variances are significantly different between groups. If it does turn out that the variances are unequal, then the standard t-test is no longer appropriate, and some variant of the test that accounts for unequal variance needs to be used, such as Welch’s t-test.

Another important assumption is independence of observations. Each measurement must correspond to a single unique individual, and not be influenced by or depend on some other measurement in the dataset. This assumption is satisfied in the present study because each patient contributes only one set of miRNA expression values, and because the observations arise from unrelated individuals.

If the data violate the normality assumption, a transformation of the miRNA expression values may be necessary. Logarithmic transformations, such as the base-10 logarithm or the natural logarithm, have commonly been used in order to correct skewness and stabilize the variance of biological expression data. While these often help normalize the distribution of expression values, it remains important to reassess normality and equality of variances after the transformation to determine whether a parametric test is still appropriate.

*Step 4:*

This analysis is conducted to assess whether there is a statistically significant relationship between two continuous variables of patient age and miRNA expression levels. Since the measurement of these two variables is on a continuous scale, the appropriate method for assessing their association is by using the Pearson Correlation Coefficient. This coefficient, denoted as r, quantifies the strength and direction of a linear relationship between two continuous variables, offering a range of -1 to +1, where values closer to either extreme represent stronger associations.

To use Pearson's correlation, there must be confirmatory evidence that the statistical assumptions needed have been met. The first assumption is that of linearity: age should relate to miRNA expression in a straight-line manner. This assumption is usually checked with a scatterplot, and its visual inspection is done for linear or non-linear trends. In case the scatterplot showed a curvature or other non-linear pattern, Pearson's correlation may not be appropriate.

The second assumption is that of normality for both variables. For Pearson's correlation to be valid, the distribution of age values and distribution of miRNA expression values should each be approximately normal. This condition is generally checked with normality tests such as the Shapiro-Wilk test, together with visual tools like a histogram or Q-Q plot. If strong deviations from normality are present, the sampling distribution of the correlation coefficient may become biased.

In case any of these necessary assumptions are violated, mathematical transformations can be performed on the data to correct it. Non-linearity or non-normality may sometimes be reduced by logarithmic, square-root, or inverse transformations in either or both variables. Such transformations may symmetrize the distributions or linearize the relation between the variables and thus permit a more appropriate application of Pearson's correlation.

If the assumptions are still not met following such adjustments then Pearson's correlation is no longer appropriate, and one needs a nonparametric alternative: this would be Spearman's Rank Correlation, abbreviated as ρ serves as the appropriate method. Unlike Pearson's correlation, Spearman's correlation does not require linearity or normality because it assesses the monotonic relationship between two ranked variables, not their actual numerical values. Thus, this makes it robust against outliers, skewed distributions, and non-linear but monotonic associations.

*Survival Analysis:*

To determine whether hsa-mir-19b-2 expression is associated with overall survival in patients with DLBCL, the continuous expression values were dichotomized at the median value of 304.094. Those patients whose expression values were greater than or equal to the median were categorized as the High Expression group, and those whose values fell below the median were categorized as the Low Expression group. Prior to comparing overall survival between the two groups, the assumptions underlying survival analysis were checked.

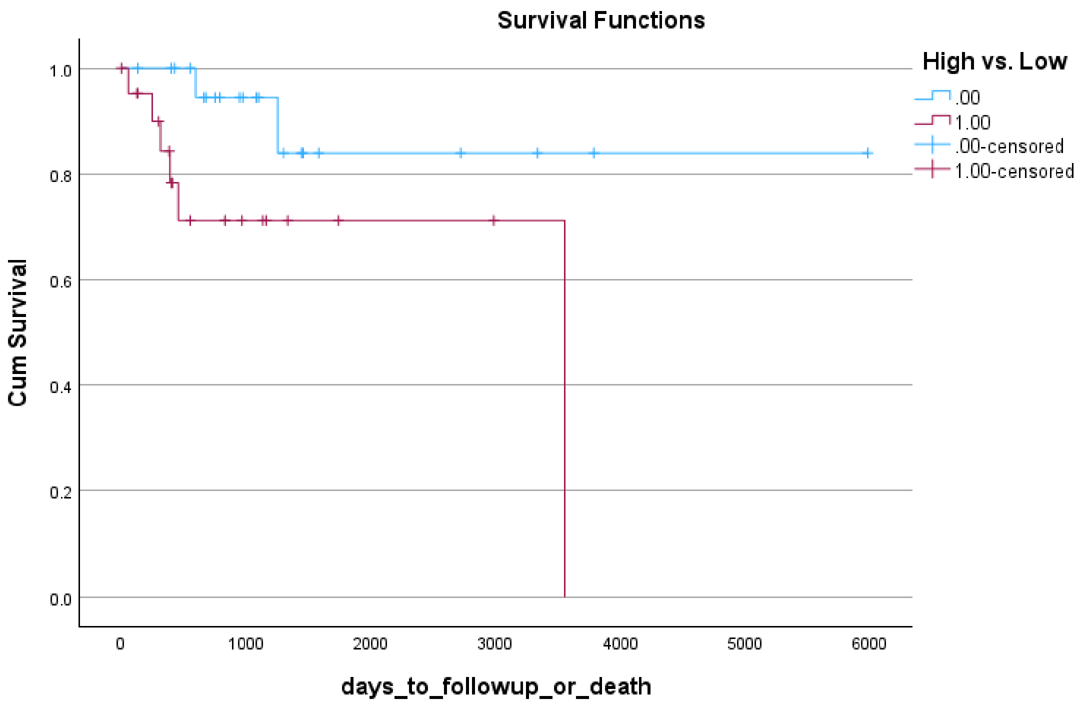


*Figure 1. Kaplan–Meier Survival Plot*

As seen in figure 1, it suggests that there was no considerable crossing of the two survival curves, implying that the assumption of proportional hazards could be reasonably met for the Log-Rank test.

Kaplan–Meier survival curves plot the overall survival for the two groups. The graphical comparison showed that the Low Expression group tended to have longer survival compared to the High Expression group.

**Overall Comparisons**

| | Chi-Square | df | Sig. |
|---|---|---|---|
| Log Rank (Mantel-Cox) | 4.417 | 1 | .036 |

Test of equality of survival distributions for the different levels of High vs. Low.

*Figure 2. Log-Rank Test Table*

This was confirmed by the Log-Rank test (Mantel–Cox) as seen in figure 2, $\chi^2 = 4.417$, df = 1, p = 0.036. Since the p-value is less than $\alpha = 0.05$, we reject the null hypothesis and conclude that there is a significant difference in survival across the two expression groups.

**Means and Medians for Survival Time**

| | Mean[a] | | | | Median | | | |
|---|---|---|---|---|---|---|---|---|
| | | | 95% Confidence Interval | | | | 95% Confidence Interval | |
| High vs. Low | Estimate | Std. Error | Lower Bound | Upper Bound | Estimate | Std. Error | Lower Bound | Upper Bound |
| .00 | 5184.685 | 536.332 | 4133.474 | 6235.896 | . | . | . | . |
| 1.00 | 2618.650 | 392.992 | 1848.385 | 3388.914 | 3553.000 | .000 | . | . |
| Overall | 4136.211 | 629.114 | 2903.147 | 5369.276 | . | . | . | . |

a. Estimation is limited to the largest survival time if it is censored.

*Figure 3. Means and Medians for Survival Time*

Summary statistics seen in figure 3 reinforce this observation. Mean survival time in the Low Expression group was 5184.7 days, 95% CI: 4133.5–6235.9, and 2618.7 days in the High Expression group, 95% CI: 1848.4–3388.9. Median survival times could not be estimated for either group, as a large portion of patients were censored prior to median survival time.

In summary, the overall results show that lower expression of hsa-mir-19b-2 was related to significantly longer survival, and this miRNA might have prognostic relevance in DLBCL.